PROJECT PROPOSAL

NumIQ: University-Specific AI Academic Assistant with Offline Support

1. Problem Statement

University students often struggle to find reliable academic assistance tailored to their university's curriculum. General-purpose AI chatbots may provide irrelevant or unsupported answers and may not recognize university faculty members or understand university-specific academic materials.

Students also need help understanding lectures notes, preparing for quizzes, and accessing academic information when an internet connection is unavailable. Therefore, a university-specific AI academic assistant is needed to provide curriculum-based answers, offline support, and other useful academic features.

2. Proposed Solution

NumIQ is a university-specific AI academic assistant designed initial for the National University of Modern Languages (NUML). It will provide students with conversational academic assistance based on university-approved course outlines, syllabi, lecture notes, and other relevant educational materials.

The system will support online and offline AI functionality through online AI services and a locally running open-source AI model. Students will be able to upload academic documents, as questions, generate practice quizzes, and search for verified faculty information.

The system will also include a moderation mechanism to flag potentially abusive or threatening content directed at university faculty or management for authorized administrative review.

NumIQ will be developed using free and open-source technologies whenever possible, with both web and mobile applications planned.

3. Main Features

1. **University-Specific AI Chatbot:** A conversational assistant designed to answer students' academic questions.
2. **Online and Offline AI Support:** Online AI functionality and a locally running open-source model for supported tasks without an internet connection.
3. **Curriculum-Based Academic Assistance:** Answers based on available university course outlines, syllabi, lecture notes, and approved academic materials.
4. **Faculty and Management Directory:** A searchable database containing faculty and management names, departments, and designations.
5. **Intelligent Faculty Identification:** Identifies possible faculty matches when a student mentions a common or incomplete name and asks for clarification when multiple people match.
6. **Conversation Moderation and Admin Alerts:** Flags potentially abusive, threatening, or harassing content targeting faculty or university management for human review. Legitimate criticism will not automatically be treated as misconduct.
7. **Upload Notes and Ask Questions:** Allows students to upload supported academic documents and ask questions, request explanations, or generate summaries from their content.
8. **AI Quiz and Practice Question Generator:** Generates multiple-choice, true/false, and short-answer questions from course topics or uploaded notes, with answers and explanations.
9. **Source-Based Answers and Out-of-Scope Handling:** Identifies relevant source documents or sections when available and acknowledges when verified information is insufficient.

4. Technology Stack

- **Web Frontend:** React.js, Vite, Tailwind CSS, and shadcn/ui.
- **Mobile Application Development:** React Native (JavaScript) or Flutter (Dart) for developing the mobile application.
- **Backend:** Node.js and Express.js
- **Database:** MongoDB Community Edition for application data, faculty records, document metadata, conversation records, quiz results, and moderation reports.
- **AI Integration:** Ollama with a suitable open-source language model for local AI functionality.
- **Document-Based AI:** Retrieval-Augmented Generation (RAG), local embeddings, and vector search for retrieving relevant academic information.
- **Document Processing:** Libraries for extracting content from supported PDF, DOCX, PPTX, and text files.
- **Development Tools:** Git and GitHub for version control, with suitable authentication and file-upload libraries.
- **Administrative Alerts:** API-based notifications or Socket.IO if real-time alerts are required.

5. Project Feasibility

NumIQ is feasible as an undergraduate final-year project because it can be developed using free and open-source technologies. The team already has experience with React.js, while Node.js, MongoDB, Flutter, and Dart can be developed further during the project.

Academic materials and faculty records will be collected and maintained for the project rather than relying on an existing university portal or database. The AI model will be selected according to available hardware, and a lightweight model will be tested early to verify offline performance.

The mobile application will use the same backend and academic data as the web application wherever practical. The team will choose either React

Native or Flutter for the actual mobile implementation to keep development manageable.

No paid AI API or paid hosting service will be mandatory for the core demonstration.

6. Project Scope

The project will focus on university-specific academic assistance, document-based question answering, quiz generation, faculty identification, online and offline AI functionality, and responsible moderation.

It will include a web application and a mobile application. It will not depend on integration with a centralized university student portal or unavailable university databases. Automated class schedules, assignment deadlines, examination timetables, and calendar integration are outside the project scope.

Offline functionality will be limited to the tasks supported by the locally installed model and locally available academic resources. Suitable hardware will be required to run the local AI model.

7. Expected Outcome

The expected outcome is a functional university-specific AI academic assistant that provide students with relevant academic information, document-based question answering, quiz generation, and faculty identification.

The system will offer web and mobile access, support selected academic tasks offline, provide source-based answers when possible, and include an administrative review mechanism for flagged conversations.

8. Conclusion

NumIQ aims to provide a practical and accessible academic assistant tailored to NUML students. By combining AI-powered academic support, locally available AI functionality, document procession, faculty information, and web and mobile applications, the project seeks to improve students' access to academic assistance.

Its reliance on free and open-source technologies makes it suitable for an undergraduate final-year project while leaving room for future improvements.